Test Report

Applicant	:	Zhejiang Mumama Toys CO.,Ltd
Address	:	No.96, Zhongshan West Road, Fenghuangshan Street, Yunhe County, Lishui City, Zhejiang Province
Manufacturer	:	Zhejiang Mumama Toys CO.,Ltd
Address	:	No.96, Zhongshan West Road, Fenghuangshan Street, Yunhe County, Lishui City, Zhejiang Province

The following sample(s) was /were submitted and identified on behalf of the clients as：

Sample Name	:	Wooden toys (track class)
Trade marks	:	Wood mom
Sample Model	:	MMM-166, MMM-167, MMM-172, MMM-173, MMM-128, MMM-112, MMM-122, MMM-111, MMM-121, MMM-117
Sample Received Date	:	Jan. 20, 2024
Testing Period	:	Jan. 20, 2024 To Jan. 25, 2024
Test Requested	:	Selected test (s) in the selected parts as requested by client.
Test Method	:	EN 71-1:2014+A1:2018, EN 71-2:2020, EN 71-3:2019+A1:2021
Test Result	:	Please refer to next page(s).
Conclusion	:	**PASS** (Based on test results)

Signed for and on behalf of

Tony Qian/ Technical Director

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Guangdong KEYS Testing Technology Co., Ltd.
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Test Result:

1. Mechanical and Physical Test

As per European standard on safety of toys EN 71-1:2014+A1:2018,

Applicant's specified age group for testing: 3 + years old.

Clause	Testing items	Assessment
4	General requirements	P
4.1	Material cleanliness	P
4.2	Assembly	P
4.3	Flexible plastic sheeting	NA
4.4	Toy bags	NA
4.5	Glass	NA
4.6	Expanding materials	NA
4.7	Edges	P
4.8	Points and metallic, wires	NA
4.9	Protruding parts	P
4.10	Parts moving against each other	P
4.10.1	Folding and sliding mechanisms	NA
4.10.2	Driving mechanisms	NA
4.10.3	Hinges	NA
4.10.4	Springs	NA
4.11	Mouth-actuated toys and other toys intended to be put in the mouth	NA
4.12	Balloons	NA

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Clause	Testing items	Assessment
4.13	Cords of toy kites and other flying toys	NA
4.14	Enclosures	--
4.14.1	Toys which a child can enter	NA
4.14.2	Masks and helmets	NA
4.15	Toys intended to bear the mass of a child	--
4.15.1	Toys propelled by a child or by other means	NA
4.15.2	Toy bicycles	NA
4.15.3	Rocking horses and similar toys	NA
4.15.4	Toys not propelled by a child	NA
4.15.5	Toy scooters	NA
4.16	Heavy immobile toys	NA
4.17	Projectiles	--
4.17.1	General	NA
4.17.2	Projectile toys without stored energy	NA
4.17.3	Projectile toys with stored energy	NA
4.17.4	Bows and arrows	NA
4.18	Aquatic toys and inflatable toys	NA
4.19	Percussion caps specifically designed for use in toys and toys using percussion caps	NA
4.20	Acoustics	NA

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Clause	Testing items	Assessment
4.21	Toys containing a non-electrical heat source	NA
4.22	Small balls	NA
4.23	Magnets	P
4.23.1	General	NA
4.23.2	Toys other than magnetic/electrical experimental sets intended for children over 8 years	NA
4.23.3	Magnetic/electrical experimental sets intended for children over 8 years	NA
4.24	Yo-yo balls	NA
4.25	Toys attached to food	NA
5	Toys intended for children under 36 months	--
5.1	General requirements	NA
5.2	Soft-filled toys and soft-filled parts of a toy	P
5.3	Plastic sheeting	NA
5.4	Cords, chains and electrical cables in toys	NA
5.5	Liquid-filled toys	NA
5.6	Speed limitation of electrically-driven ride-on toys	NA
5.7	Glass and porcelain	NA
5.8	Shape and size of certain toys	NA
5.9	Toys comprising monofilament fibres	NA
5.10	Small balls	NA

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Clause	Testing items	Assessment
5.11	Play figures	NA
5.12	Hemispheric-shaped toys	NA
5.13	Suction cups	NA
5.14	Straps intended to be worn fully or partially around the neck	NA
5.15	Sledges with cords for pulling	NA
6	Packaging	NA
7	Warnings, markings and instructions for use	--
7.1	General	NA
7.2	Toys not intended for children under 36 months	NA
7.3	Latex balloons	NA
7.4	Aquatic toys	NA
7.5	Functional toys	NA
7.6	Hazardous sharp functional edges and points	--
7.7	Projectiles	NA
7.7.1	Toys with projectiles which are able to discharge an object other than that provided with the toy	NA
7.7.2	Toys capable of discharging a projectile with a kinetic energy greater than 0.08 j	NA
7.8	Imitation protective masks and helmets	NA
7.9	Toy kites	NA
7.10	Roller skates, inline skates, skateboards and certain other ride-on toys	--

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Clause	Testing items	Assessment
7.10.1	Roller skates, inline skates and skateboards	NA
7.10.2	Ride-on toys without a braking device	NA
7.10.3	Electrically-driven ride-on toys	NA
7.10.4	Instructions for use	NA
7.11	Toys intended to be attached to or strung across a cradle, cot, or perambulator	NA
7.12	Liquid-filled teethers	NA
7.13	Percussion caps specifically designed for use in toys	NA
7.14	Acoustics	NA
7.15	Toy bicycles	NA
7.16	Toys intended to bear the mass of a child	NA
7.17	Toys comprising monofilament fibers	NA
7.18	Toy scooters	NA
7.19	Rocking horses and similar toys	NA
7.20	Magnetic/electrical experimental sets	NA
7.21	Toys with electrical cables exceeding 300 mm in length	NA
7.22	Toys with cords or chains intended for children of 18 months and over but under 36months	NA
7.23	Toys intended to be attached to a cradle, cot or perambulator	NA
7.24	Sledges with cords for pulling	NA
7.25	Flying toys	NA

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Clause	Testing items	Assessment
7.25.1	Flying toys	NA
7.25.2	Remote controlled flying toys	NA
7.26	Improvised projectiles	NA
Remark :	1. P=Pass, 2. NA= Not Applicable, 3. F=Fail	

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Test results and description

Clause	Test Result
4.1	Materials Visual Clean, No Infestation
4.7	No sharp edges
4.9	No sharp points
8.34 Tension test for magnets	The magnet did not fall off after the test
8.35 Magnetic flux index	flux index 172.5 kG2 mm2
8.35 Magnetic flux index	flux index 107.1 kG2 mm2

2. Flammability Test

As per European standard on safety of toys EN 71-2:2020

Clause	Testing Items	Assessment
4.1	General requirements	P
4.2	Toys to be worn on the head	NA
4.3	Toy disguise costumes and toys intended to be worn by a child in play	NA
4.4	Toys intend to be entered by a child	NA
4.5	Soft-filled toys	NA
Remark:	1. P = Pass; 2. NA = Not applicable.	

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3. Migration Of Certain Elements

Test Method: As per European standard on safety of toys EN 71-3:2019 +A1:2021

Sample Description:

No.	Name
1	Yellow Wood
2	Red + Yellow + Sky Blue Paint (Mix)
3	Green + White + Blue Paint (Mix)
4	Red + Yellow + Black Plastic (Mix)
5	Orange + Dark Blue + Dark Grey Plastic (Mix)
6	Brown Wood
7	Silver Metal
8	Screw
9	Pink + Grey + Brown Paint (Mix)
10	Light Grey + Orange + Light Green Paint (Mix)

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The Result:

Migration test

Test Item(s)	MDL	Unit	Result				
			No.1	No.2	No.3	No.4	No.5
Soluble Aluminium (Al)	50	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Antimony(Sb)	1.0	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Arsenic(As)	0.5	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Barium(Ba)	50	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Boron(B)	50	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Cadmium(Cd)	0.3	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Chromium(III)(Cr III)	5	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Chromium(VI) (Cr VI)	0.005	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Cobalt(Co)	0.5	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Copper(Cu)	50	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Lead(Pb)	0.5	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Manganese(Mn)	50	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Mercury(Hg)	0.5	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Nickel(Ni)	10	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Selenium(Se)	0.08	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Strontium(Sr)	5.0	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Tin(Sn)	50	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Organic tin	0.5	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Zinc(Zn)	0.08	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.

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Test Item(s)	MDL	Unit	Result				
			No.6	No.7	No.8	No.9	No.10
Soluble Aluminium (Al)	50	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Antimony(Sb)	1.0	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Arsenic(As)	0.5	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Barium(Ba)	50	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Boron(B)	50	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Cadmium(Cd)	0.3	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Chromium(III)(Cr III)	5	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Chromium(VI) (Cr VI)	0.005	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Cobalt(Co)	0.5	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Copper(Cu)	50	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Lead(Pb)	0.5	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Manganese(Mn)	50	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Mercury(Hg)	0.5	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Nickel(Ni)	10	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Selenium(Se)	0.08	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Strontium(Sr)	5.0	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Tin(Sn)	50	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Organic tin	0.5	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.
Soluble Zinc(Zn)	0.08	mg/kg	N.D.	N.D.	N.D.	N.D.	N.D.

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Organic tin test

Test Item(s)	Unit	MDL	Limit	Result				
				No.1	No.2	No.3	No.4	No.5
Methyltin	mg/kg	0.5	12	N.D.	N.D.	N.D.	N.D.	N.D.
Dimethyltin	mg/kg	0.5		N.D.	N.D.	N.D.	N.D.	N.D.
Butyltin	mg/kg	0.5		N.D.	N.D.	N.D.	N.D.	N.D.
Tributyltin	mg/kg	0.5		N.D.	N.D.	N.D.	N.D.	N.D.
Monooctyltin	mg/kg	0.5		N.D.	N.D.	N.D.	N.D.	N.D.
Dioctyltin	mg/kg	0.5		N.D.	N.D.	N.D.	N.D.	N.D.
Dibutyltin	mg/kg	0.5		N.D.	N.D.	N.D.	N.D.	N.D.
Dipropyltin	mg/kg	0.5		N.D.	N.D.	N.D.	N.D.	N.D.
Tetrabutyltin	mg/kg	0.5		N.D.	N.D.	N.D.	N.D.	N.D.
Diphenyltin	mg/kg	0.5		N.D.	N.D.	N.D.	N.D.	N.D.
Triphenyltin	mg/kg	0.5		N.D.	N.D.	N.D.	N.D.	N.D.

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Test Item(s)	Unit	MDL	Limit	Result				
				No.6	No.7	No.8	No.9	No.10
Methyltin	mg/kg	0.5	12	N.D.	N.D.	N.D.	N.D.	N.D.
Dimethyltin	mg/kg	0.5		N.D.	N.D.	N.D.	N.D.	N.D.
Butyltin	mg/kg	0.5		N.D.	N.D.	N.D.	N.D.	N.D.
Tributyltin	mg/kg	0.5		N.D.	N.D.	N.D.	N.D.	N.D.
Monooctyltin	mg/kg	0.5		N.D.	N.D.	N.D.	N.D.	N.D.
Dioctyltin	mg/kg	0.5		N.D.	N.D.	N.D.	N.D.	N.D.
Dibutyltin	mg/kg	0.5		N.D.	N.D.	N.D.	N.D.	N.D.
Dipropyltin	mg/kg	0.5		N.D.	N.D.	N.D.	N.D.	N.D.
Tetrabutyltin	mg/kg	0.5		N.D.	N.D.	N.D.	N.D.	N.D.
Diphenyltin	mg/kg	0.5		N.D.	N.D.	N.D.	N.D.	N.D.
Triphenyltin	mg/kg	0.5		N.D.	N.D.	N.D.	N.D.	N.D.

Note:

1. mg/kg= ppm;
2. N.D. =Not Detected (<MDL);
3. MDL =Method Detection Limit.

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Cross-reference table for determining category

Toy Material	Category I	Category II	Category III
Coatings of paints, varnishes, lacquers, printing inks, polymers, foams and similar coatings			X
Polymeric and similar materials, including laminates, whether textile reinforced or not, but excluding other textiles			X
Paper and paper board			X
Textiles, whether natural or synthetic			X
Glass, ceramic, metallic materials			X
Wood, fibre board, hard board, bone, leather and other solid materials			X
Compressed paint tablets, materials intended to leave a trace or similar materials in solid form appearing as such in the toy (e.g. the cores of colouring pencils, chalk, crayons)	X		
Pliable modelling materials, including modelling clays and plaster	X		
Liquid paints, including finger paints, varnishes, lacquers, liquid ink in pens and similar materials in liquid form appearing as such in the toy (e.g. slimes, bubble solution)		X	
Glue sticks		X	

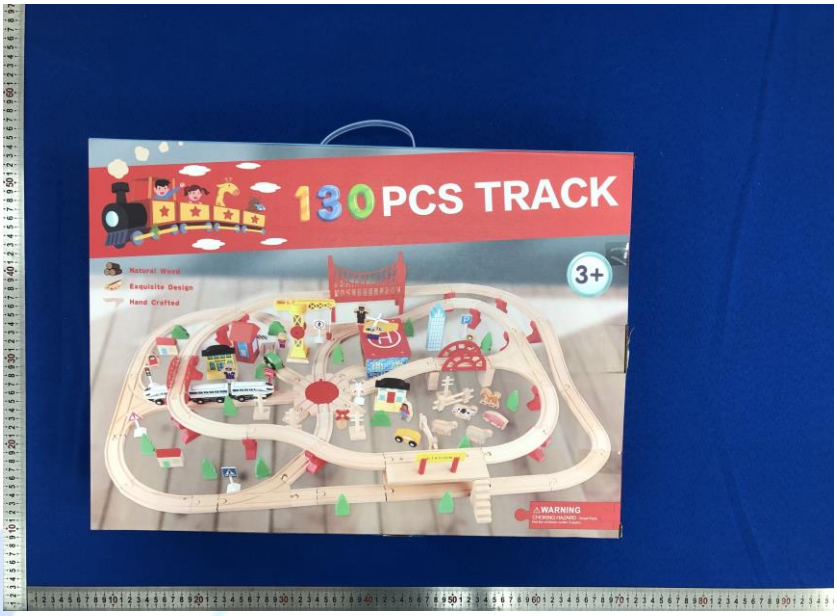
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Migration limits from toy materials

Element	Migration limit		
	Category I(mg/kg)	Category II(mg/kg)	Category III(mg/kg)
Aluminium	2250	560	28130
Antimony	45	11.3	560
Arsenic	3.8	0.9	47
Barium	1500	375	18750
Boron	1200	300	15000
Cadmium	1.3	0.3	17
Chromium (III)	37.5	9.4	460
Chromium (VI)	0.02	0.005	0.053
Cobalt	10.5	2.6	130
Copper	622.5	156	7700
Lead	2.0	0.5	23
Manganese	1 200	300	15 000
Mercury	7.5	1.9	94
Nickel	75	18.8	930
Selenium	37.5	9.4	460
Strontium	4500	1125	56000
Tin	15000	3750	180000
Organic tin	0.9	0.2	12
Zinc	3750	938	46000
Note: The migration limits for Aluminium have been amended by Commission Directive (EU) 2019/1922			

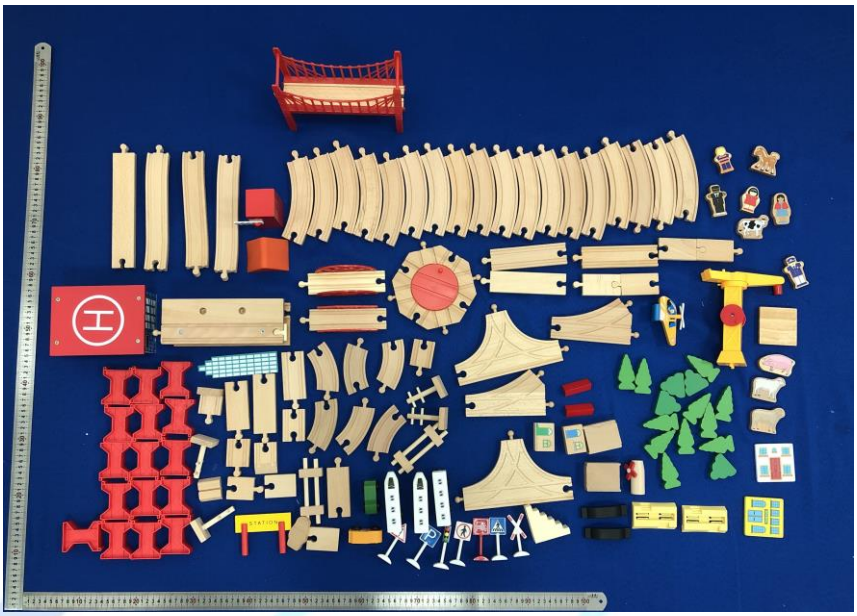
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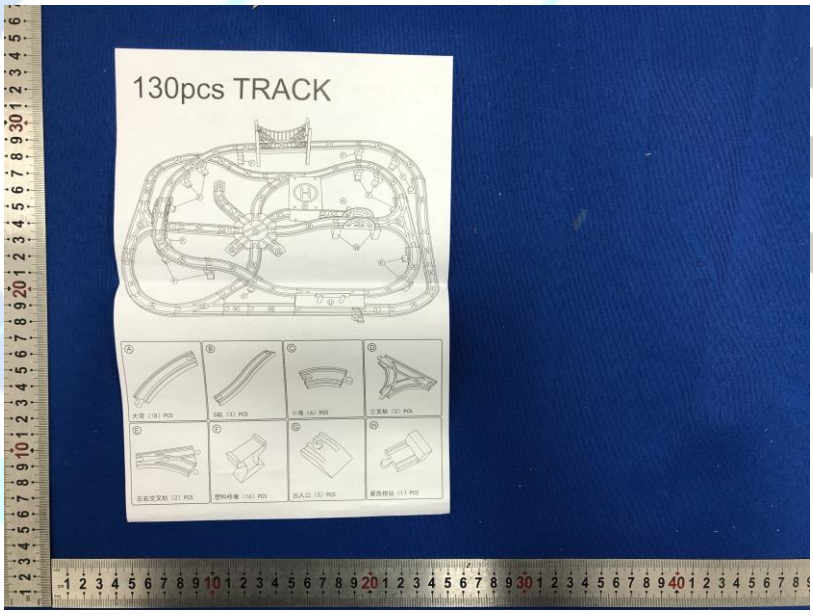
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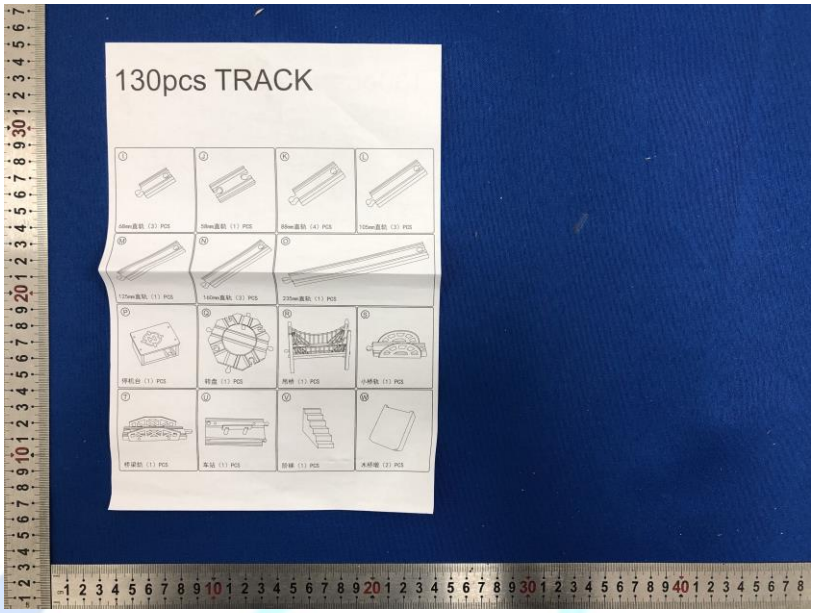


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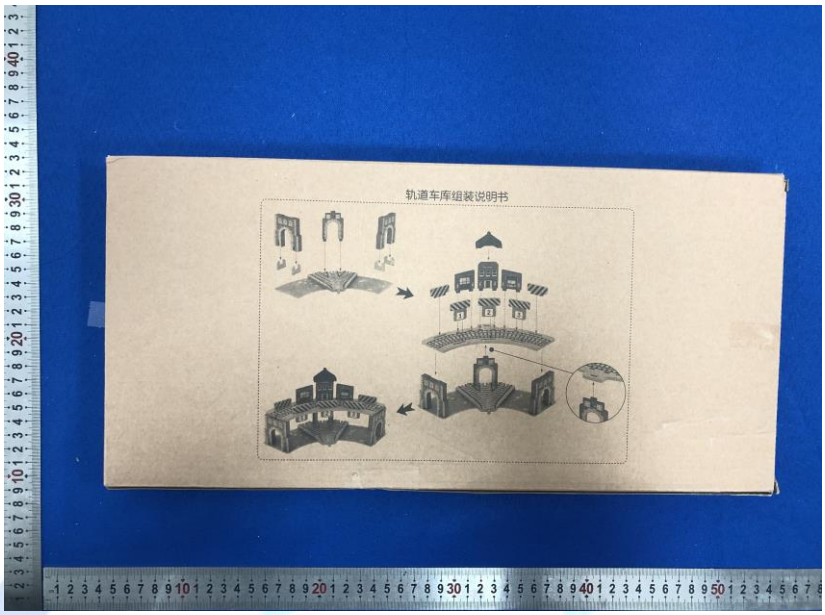


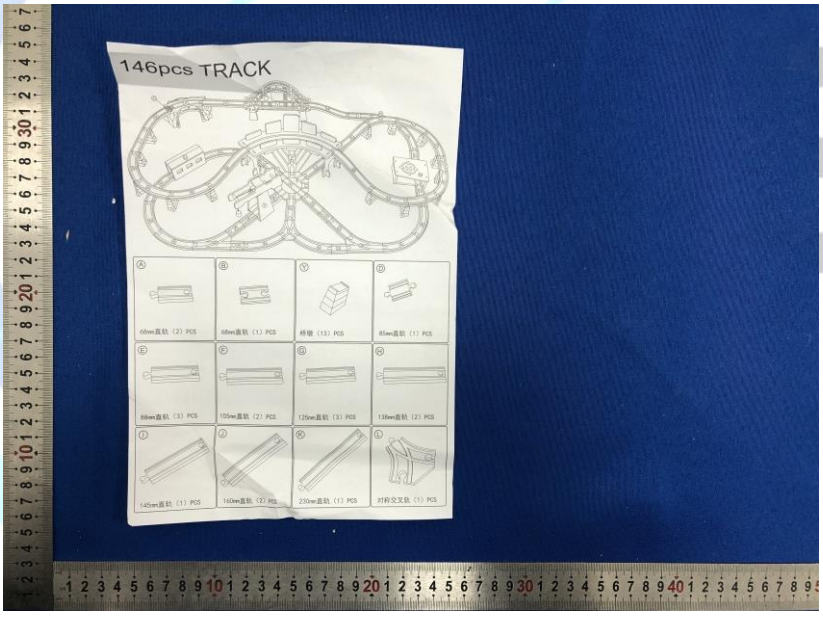
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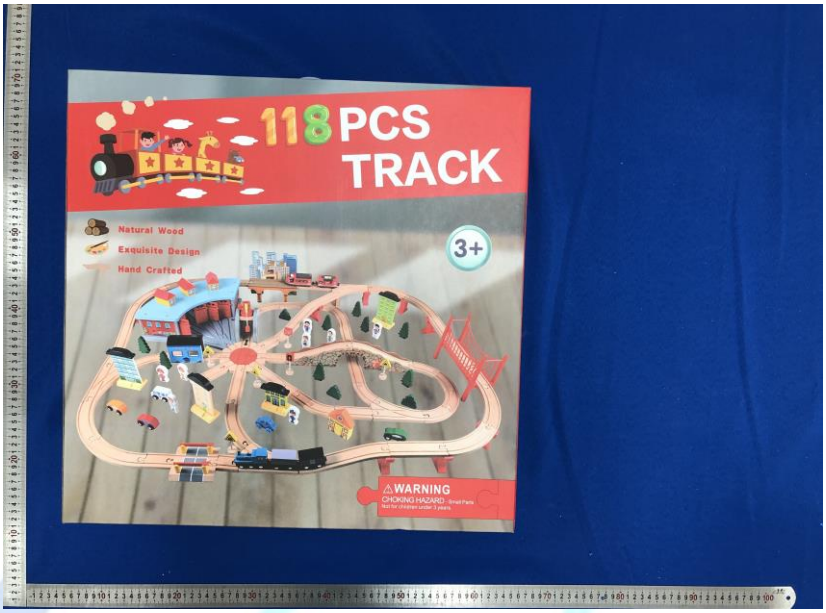


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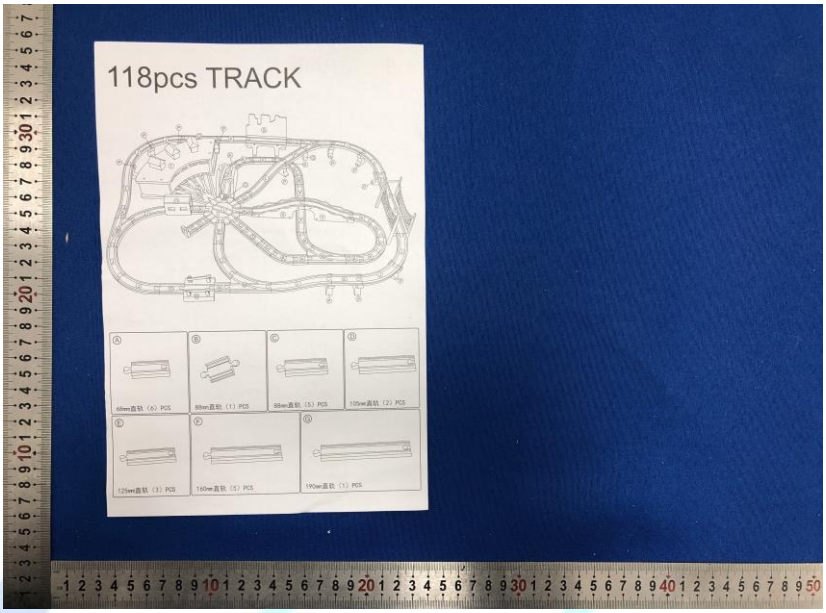


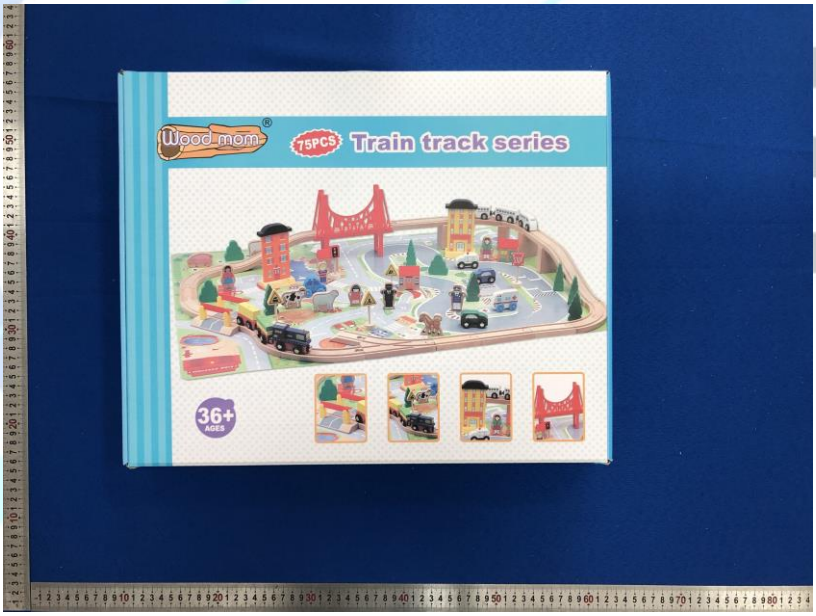
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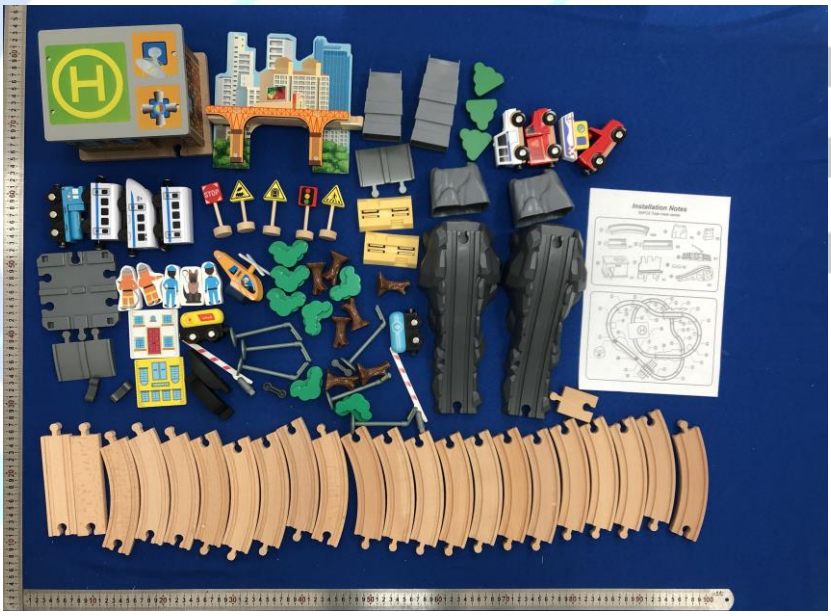
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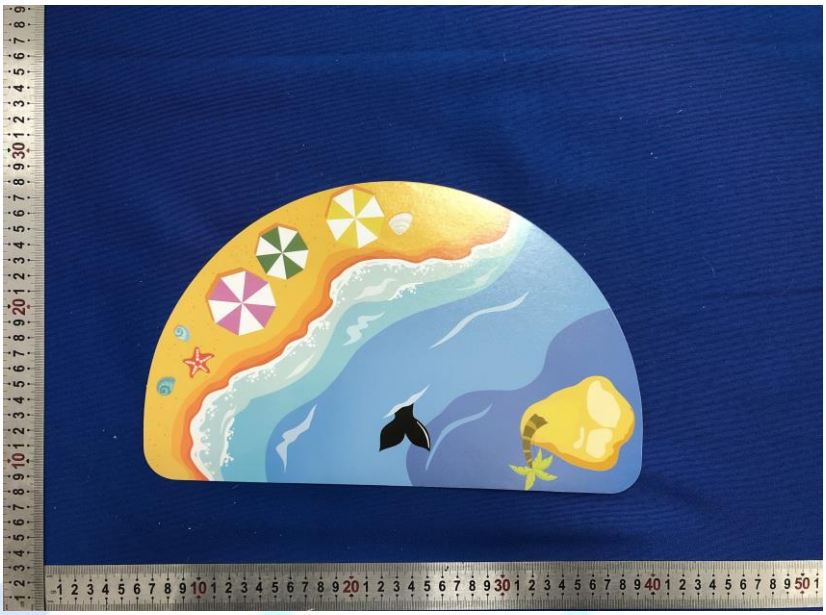


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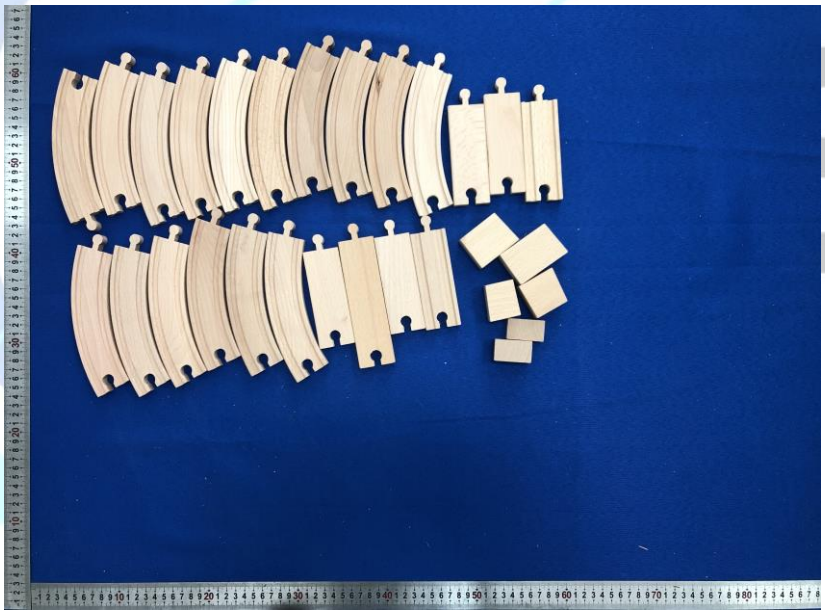
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*** End of Report ***

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